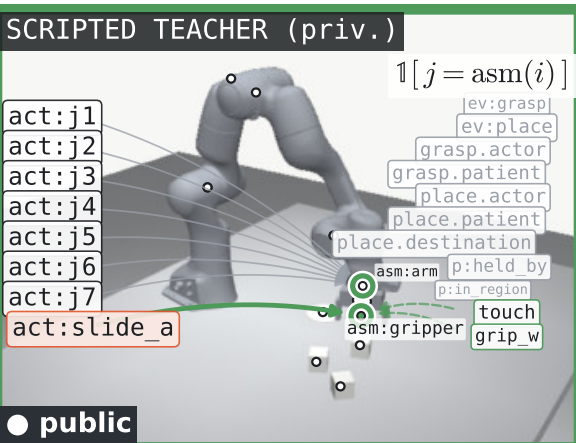
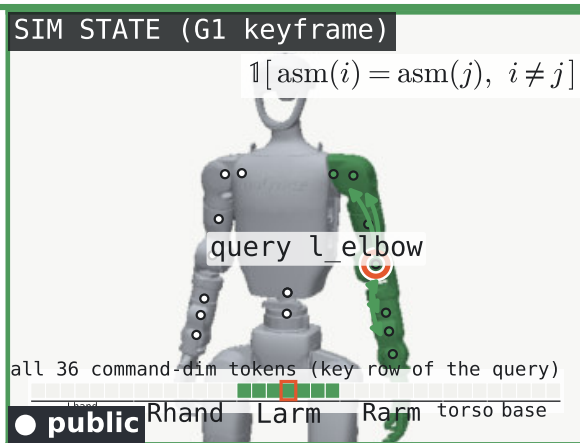


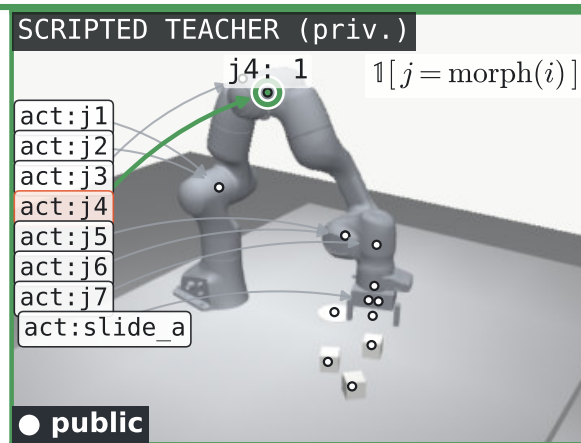
## STRUCTURE / EDGES (9 implemented)



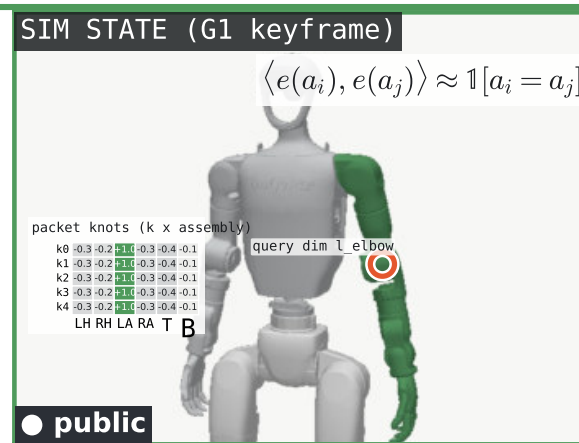
**edge.node\_in\_assembly**  
joint / sensor → its assembly



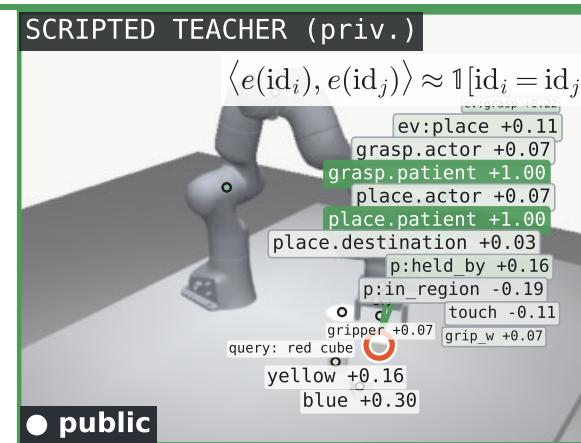
**edge.same\_assembly**  
dims of the same assembly



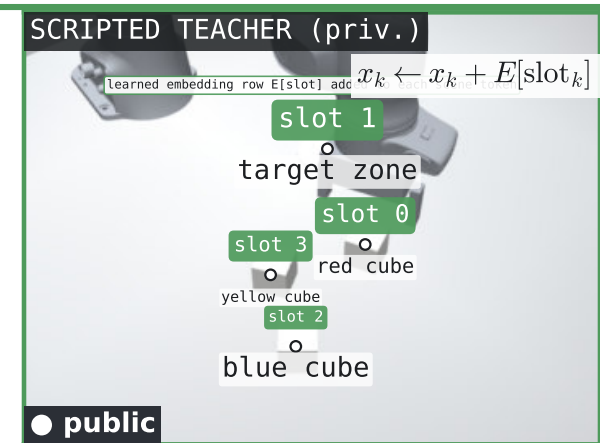
**edge.same\_node**  
action node → its own joint



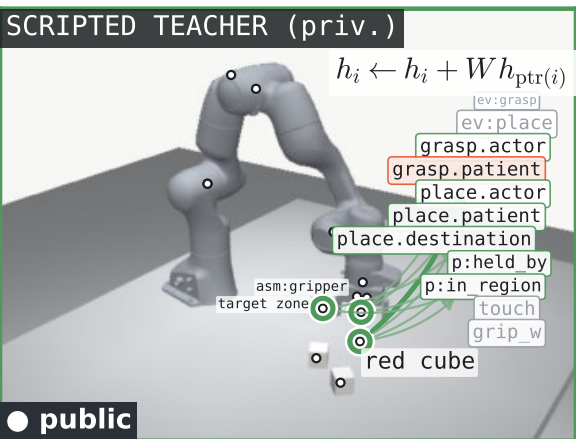
**id.same\_assembly**  
soft same-assembly id match



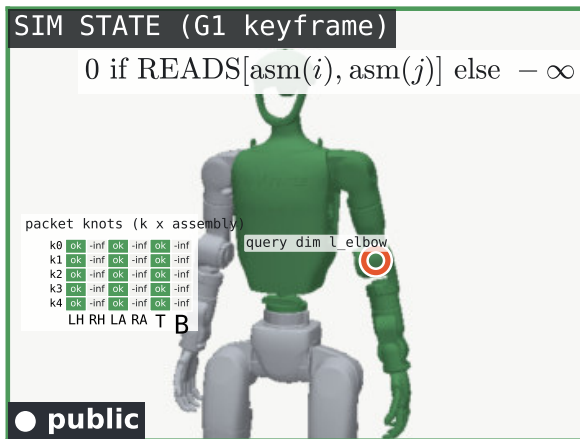
**id.same\_body**  
tokens of the same entity



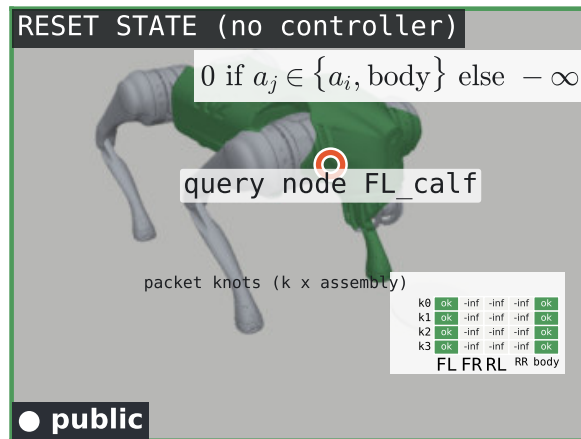
**id.slot\_handle**  
tracker-slot identity embedding



**msg.incidence**  
entity → role / predicate message

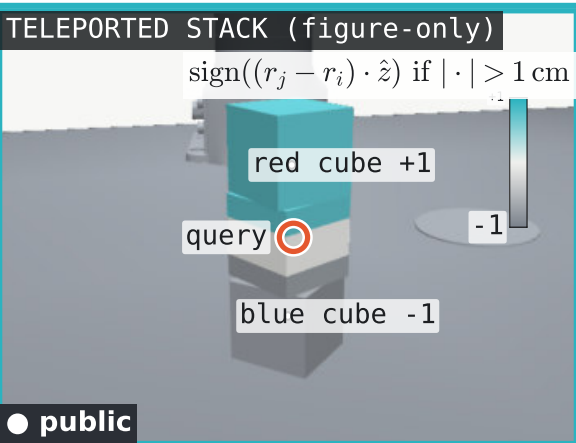


**route.assembly\_reads**  
dim reads own + neighbour knots

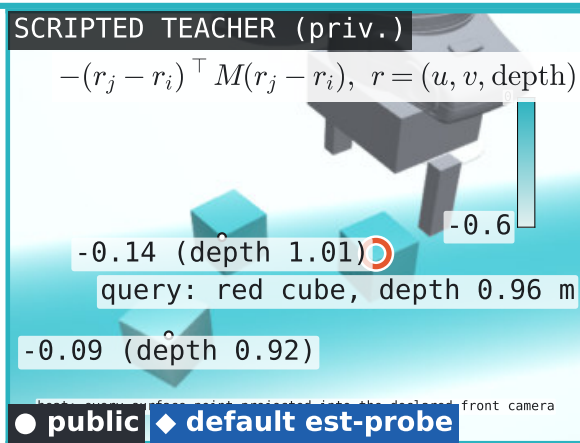


**route.own\_assembly**  
joint reads own-limb knots

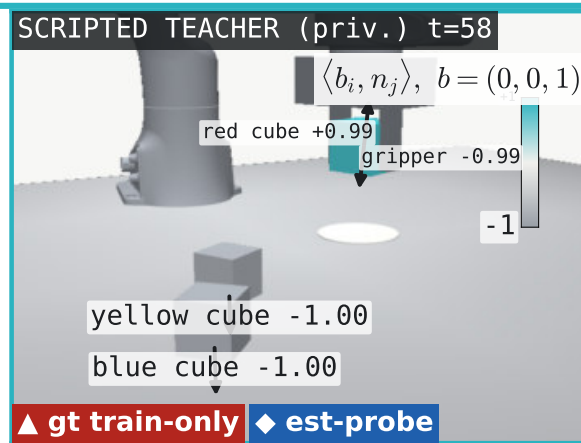
## GEOMETRY (5 implemented)



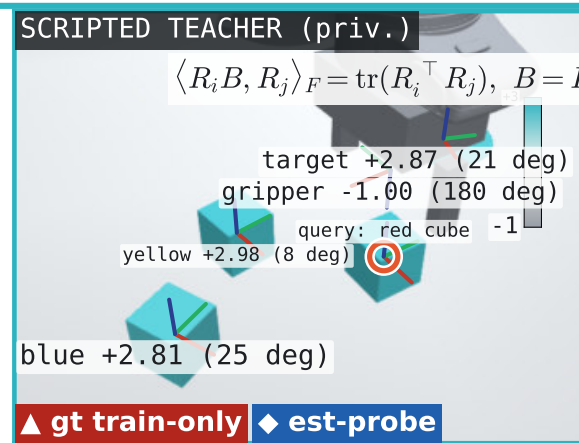
**geo.above**  
above / below along gravity



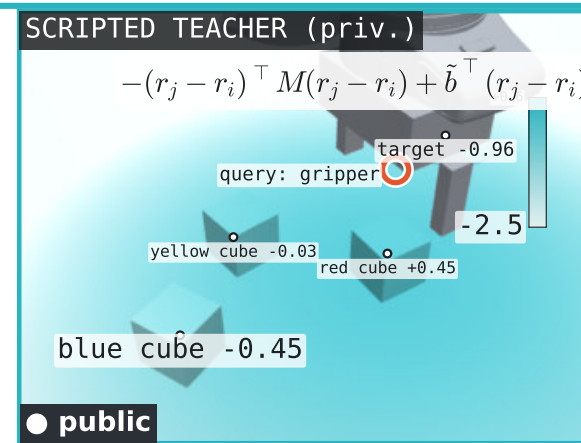
**geo.depth3d**  
PaPE on camera (u, v, depth)



**geo.normal\_align**  
direction vs contact normal

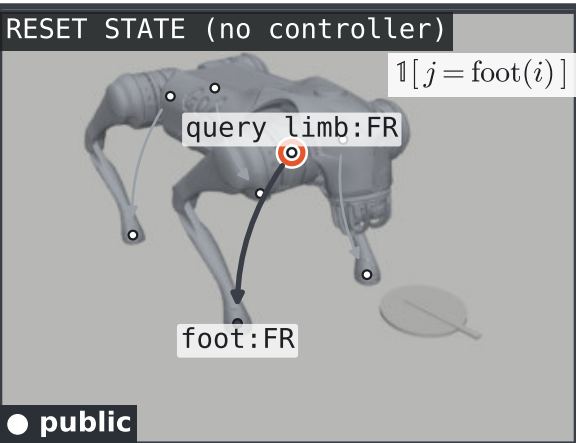


**geo.orient**  
relative rotation of frames

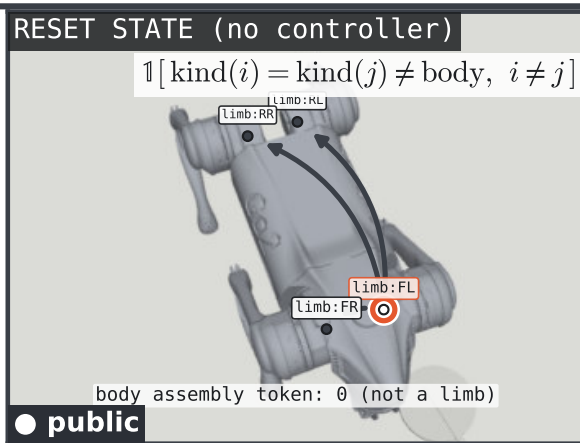


**geo.pos3d**  
PaPE relative 3-D position

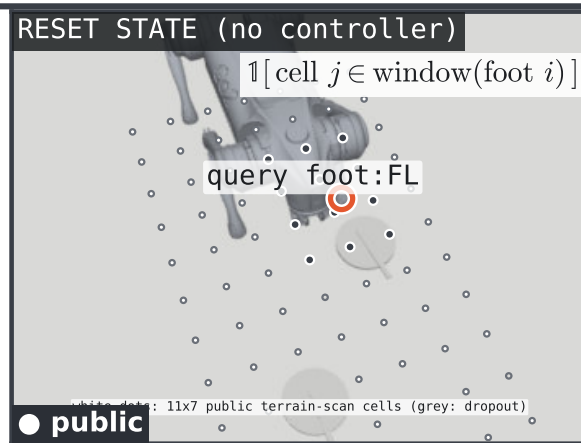
## KINEMATICS / BODY (8 implemented)



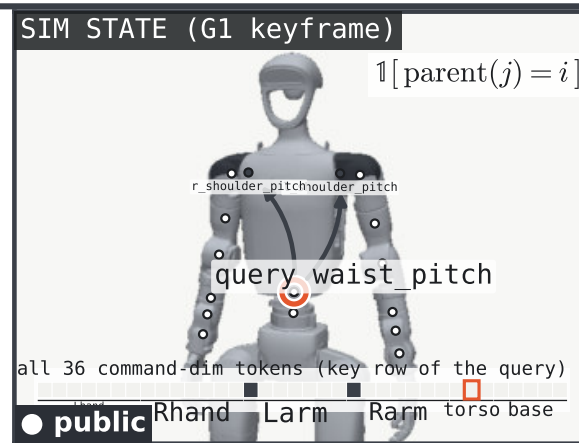
**edge.foot\_of**  
limb → its own foot



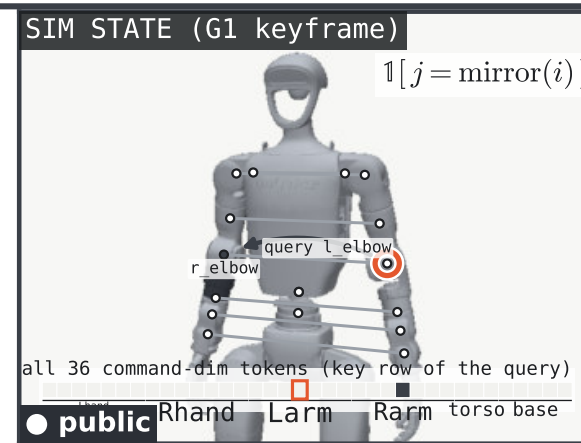
**edge.limb\_adjacent**  
limb ↔ limbs of the same kind



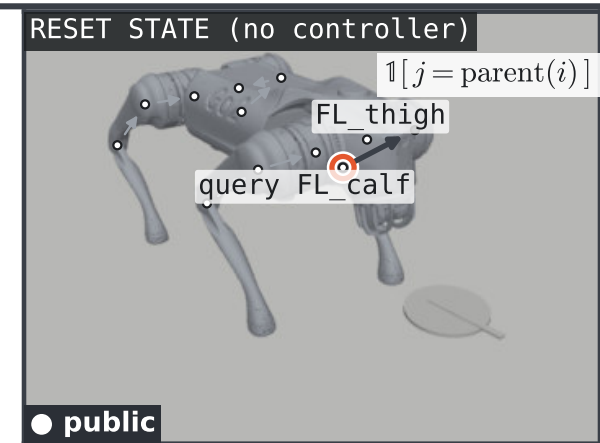
**edge.over\_cell**  
foot → landing-window cells



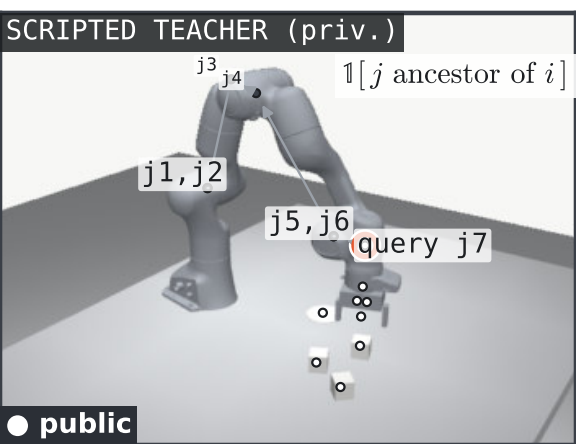
**edge.kin\_child**  
dim → kinematic children



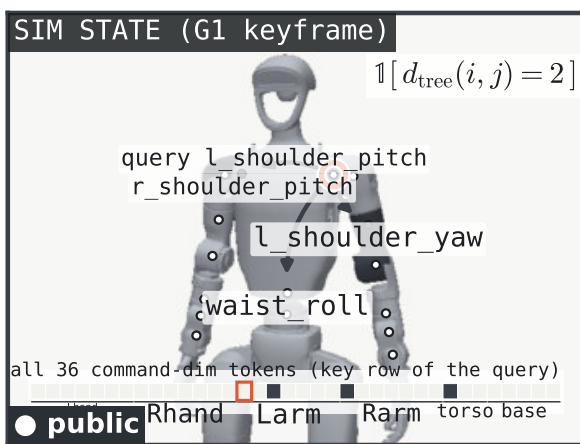
**edge.mirror**  
dim ↔ left / right homologue



**edge.kin\_parent**  
joint → kinematic parent

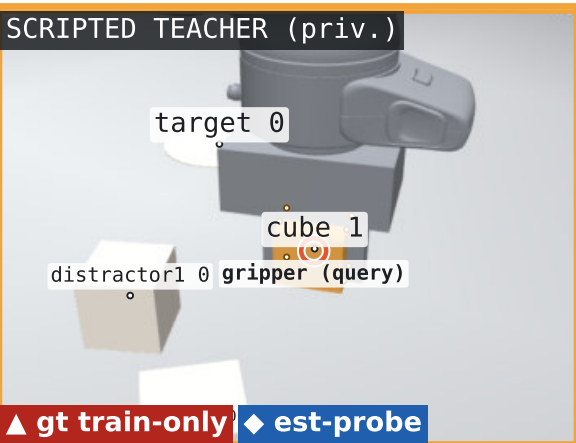


**kin.ancestor**  
joint → all chain ancestors

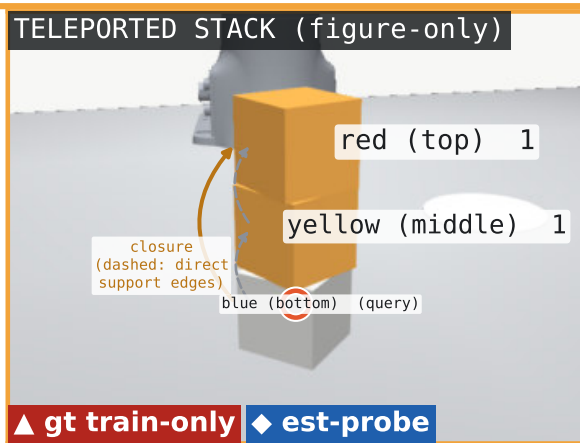


**kin.sibling**  
dims at tree distance 2

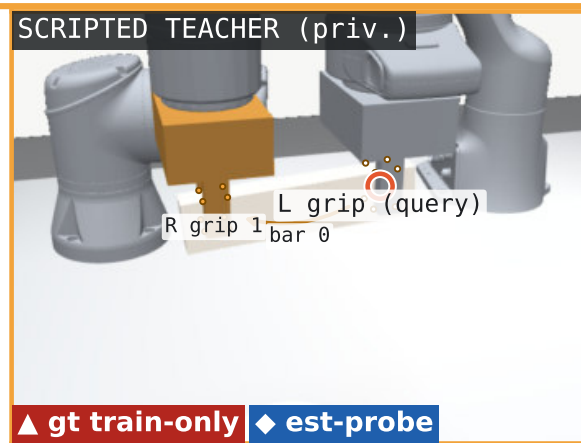
## PHYSICAL INTERACTION (5 implemented)



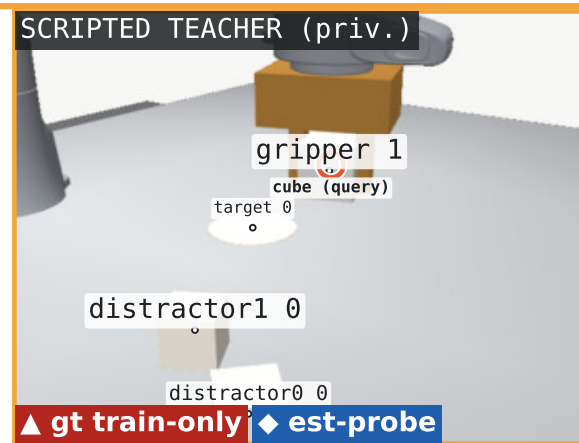
**ix.contact**  
entities in geometric contact



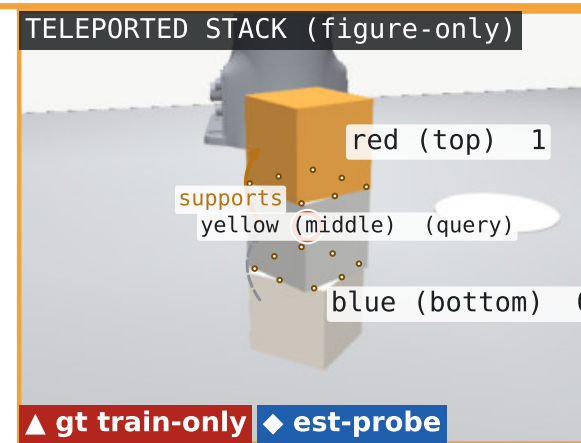
**ix.force\_flow**  
everything the query carries



**ix.handover**  
two hands on one object



**ix.held\_by**  
object held by a hand



**ix.support**  
a supports b (contact, up)